Diet and thrombosis risk: nutrients for prevention of thrombotic disease.

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An increased prothrombotic state is a major risk factor for the development of heart attacks, strokes, and venous thromboembolism. Platelet activation and aggregation play an important role in determining a prothrombotic state. Although pharmaceutical agents such as aspirin, heparin, and warfarin are able to reduce prothrombotic tendency, long-term drug treatment may produce a variety of side effects, including bleeding. Diet is generally recognized to be significantly involved in modifying the individual risk for the development of thrombotic diseases, although its influence during the treatment of these disorders is probably less important. Dietary intervention has proven effective in lowering serum lipid levels, which are otherwise essential elements in the pathogenesis of cardiovascular disease. Likewise, certain dietary components have also been proven effective in decreasing platelet activation through various mechanisms and therefore may contribute to attenuating the future risk of thrombosis. This article provides an up-to-date review of the role of nutrient and nonnutrient supplements on platelet aggregation and risk of thrombosis.